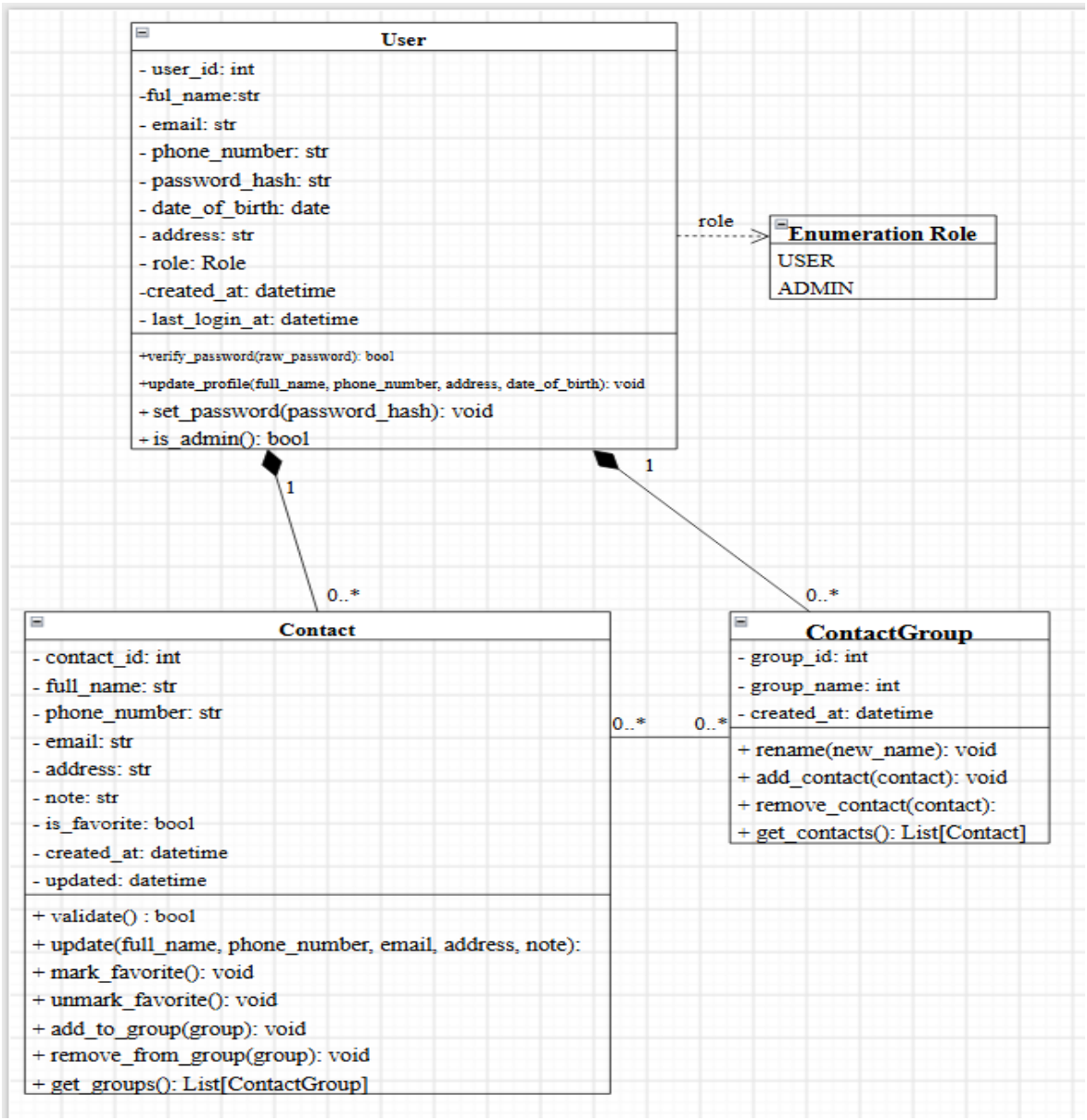
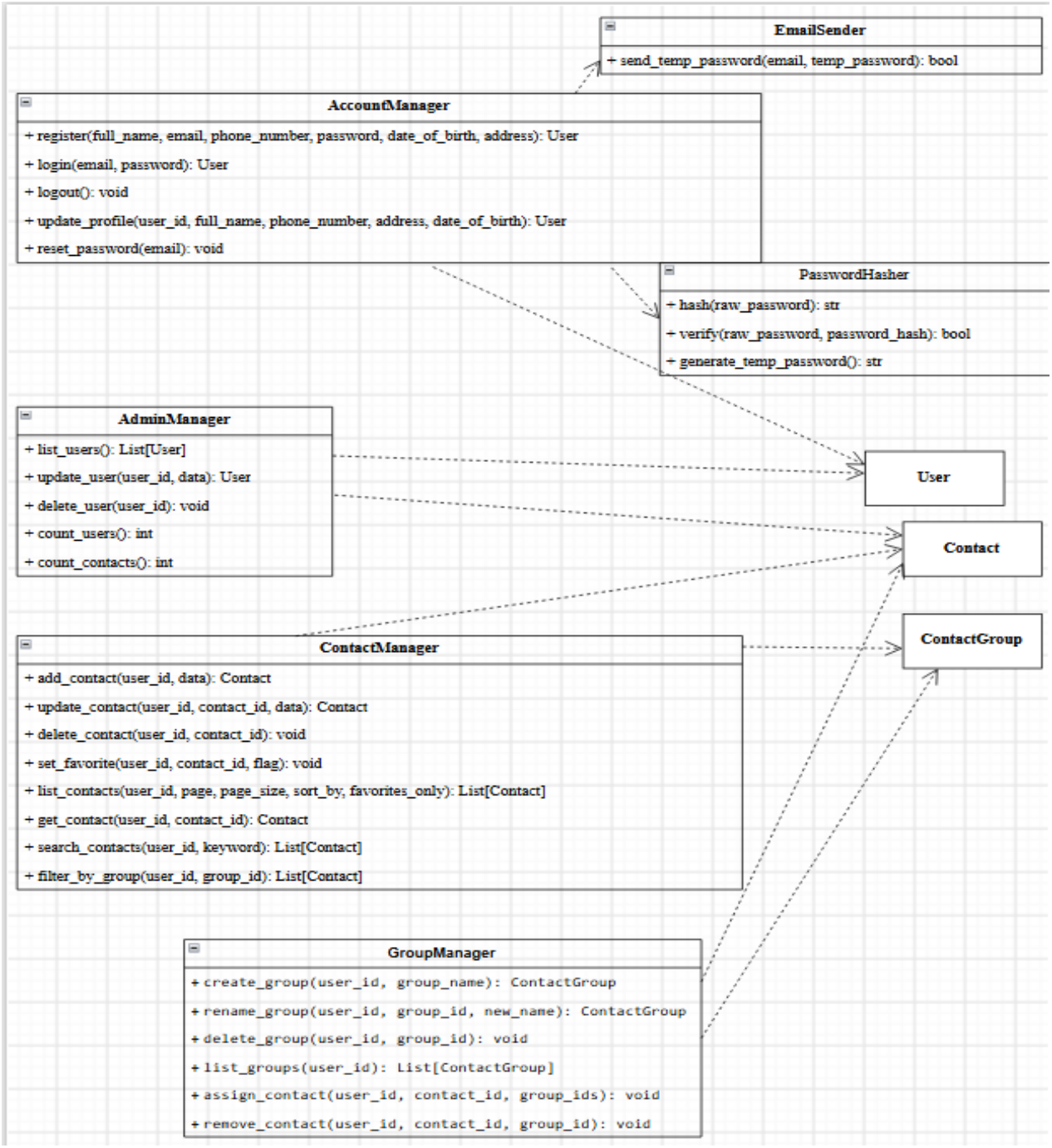
5. Class Diagram

5.1. Entity class diagram



5.2. Processing layer class diagram



6. Data Model

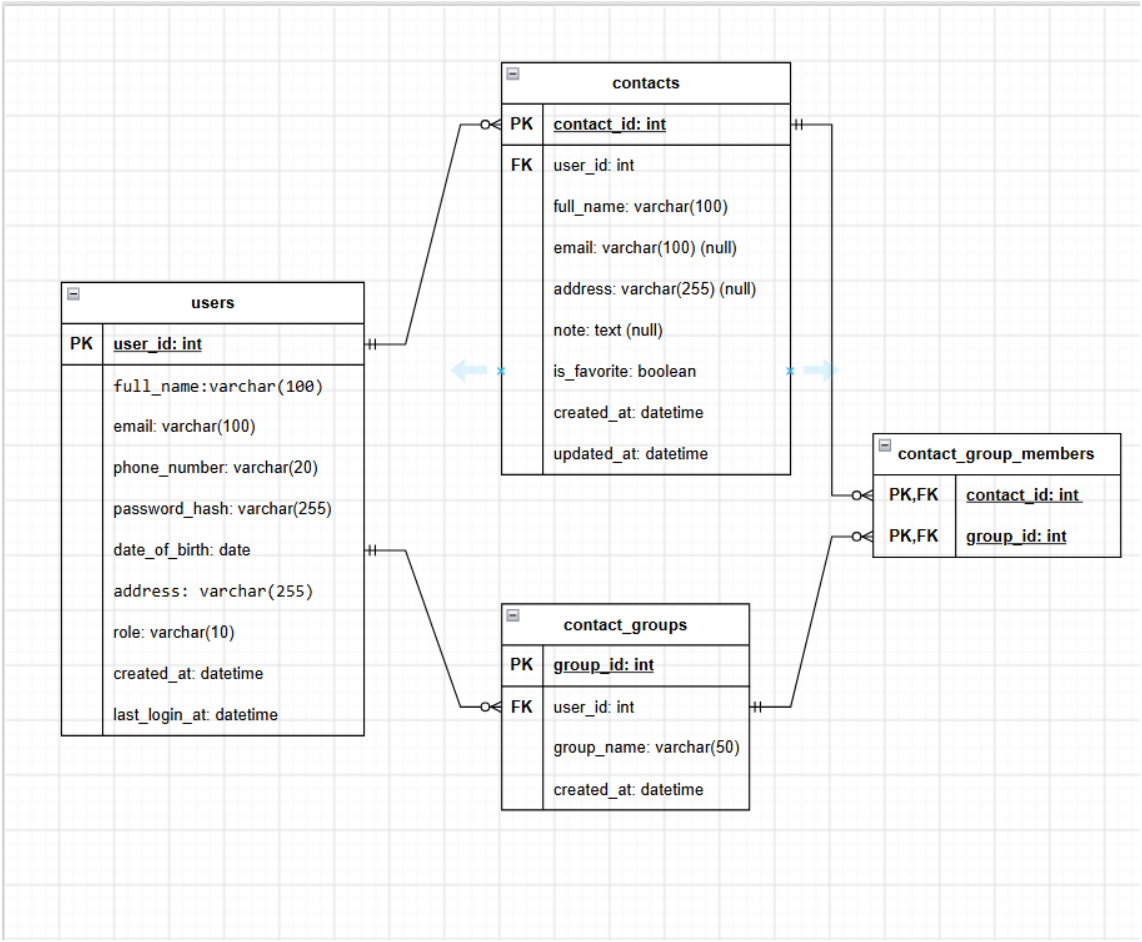
6.1. Mapping between the DFD, classes and database tables

DFD data store	Class	MySQL table	Note
D1 Users	User	users	Accounts and roles

D2 Contacts	Contact	contacts	Each user's contacts
D3 Groups	ContactGroup	contact_groups, contact_group_members	Groups and the links between contacts and groups

The many-to-many relationship between Contact and ContactGroup is implemented by the junction table contact_group_members, which belongs to the data store D3 Groups.

6.2. Entity relationship diagram (ERD)



6.3. Table specifications

The database management system is MySQL. Tables use the InnoDB engine to support transactions and foreign keys, and the utf8mb4 character set to store Vietnamese text.

Table users

DFD data store: D1 Users. Stores user and administrator accounts.

Column	Data type	Key	Constraints	Description
user_id	INT	PK	PRIMARY KEY; AUTO_INCREMENT	Primary key, auto-increment
full_name	VARCHAR(100)		NOT NULL	Full name
email	VARCHAR(100)		NOT NULL; UNIQUE	Email, used as the login name
phone_number	VARCHAR(20)		NOT NULL	Phone number
password_hash	VARCHAR(255)		NOT NULL	Hashed password
date_of_birth	DATE		NOT NULL	Date of birth
address	VARCHAR(255)		NOT NULL	Address
role	VARCHAR(10)		NOT NULL; DEFAULT 'USER'; CHECK (role IN ('USER','ADMIN'))	Role
created_at	DATETIME		NOT NULL; DEFAULT CURRENT_TIMESTAMP	Time the account was created
last_login_at	DATETIME		NULL	Most recent login time

Table contacts

DFD data store: D2 Contacts. Stores each contact in a user's personal phone book.

Column	Data type	Key	Constraints	Description
contact_id	INT	PK	PRIMARY KEY; AUTO_INCREMENT	Primary key, auto-increment

user_id	INT	FK	NOT NULL; FK → users(user_id) ON DELETE CASCADE	Owner of the contact
full_name	VARCHAR(100)		NOT NULL	Contact full name
phone_number	VARCHAR(20)		NOT NULL	Phone number
email	VARCHAR(100)		NULL	Contact email address
address	VARCHAR(255)		NULL	Address
note	TEXT		NULL	Note (optional)
is_favorite	BOOLEAN		NOT NULL; DEFAULT FALSE	Favorite flag
created_at	DATETIME		NOT NULL; DEFAULT CURRENT_TIMESTAMP	Creation time
updated_at	DATETIME		NOT NULL; DEFAULT CURRENT_TIMESTAMP	Time of the most recent update

Table contact_groups

DFD data store: D3 Groups. Stores the groups (categories) created by users, e.g. Family, Friends, Work.

Column	Data type	Key	Constraints	Description
group_id	INT	PK	PRIMARY KEY; AUTO_INCREMENT	Primary key, auto-increment
user_id	INT	FK	NOT NULL; FK → users(user_id) ON DELETE CASCADE	User who created the group
group_name	VARCHAR(50)		NOT NULL	Group name
created_at	DATETIME		NOT NULL; DEFAULT CURRENT_TIMESTAMP	Creation time

Additional unique constraint: (user_id, group_name).

Table contact_group_members

DFD data store: D3 Groups. Junction table of the many-to-many relationship between contacts and groups.

Column	Data type	Key	Constraints	Description
contact_id	INT	PK, FK	PRIMARY KEY; FK → contacts(contact_id) ON DELETE CASCADE	Part 1 of the primary key, references the contact
group_id	INT	PK, FK	PRIMARY KEY; FK → contact_groups(group_id) ON DELETE CASCADE	Part 2 of the primary key, references the group

6.4. Indexes

Primary keys and UNIQUE constraints are indexed automatically. The additional indexes support the performance requirements (search results within 0.5 seconds for 10,000 records, up to 50,000 contacts per account):

Index name	Table	Columns	Purpose
idx_contacts_user_name	contacts	(user_id, full_name)	Listing, sorting A-Z / Z-A and searching by name within one user's contacts
idx_contacts_user_phone	contacts	(user_id, phone_number)	Searching by phone number
idx_contacts_user_created	contacts	(user_id, created_at)	Sorting by creation date
idx_contacts_user_fav	contacts	(user_id, is_favorite)	Viewing the favorites list

idx_members_group	contact_group_members	(group_id)	Filtering contacts by group